

Software Engineering

Lab 1: Requirements Engineering & UML Use-Case Modelling

Problem Statement 13:

Patient Health Record Consent Management System

NAME: GAGANDEEP N

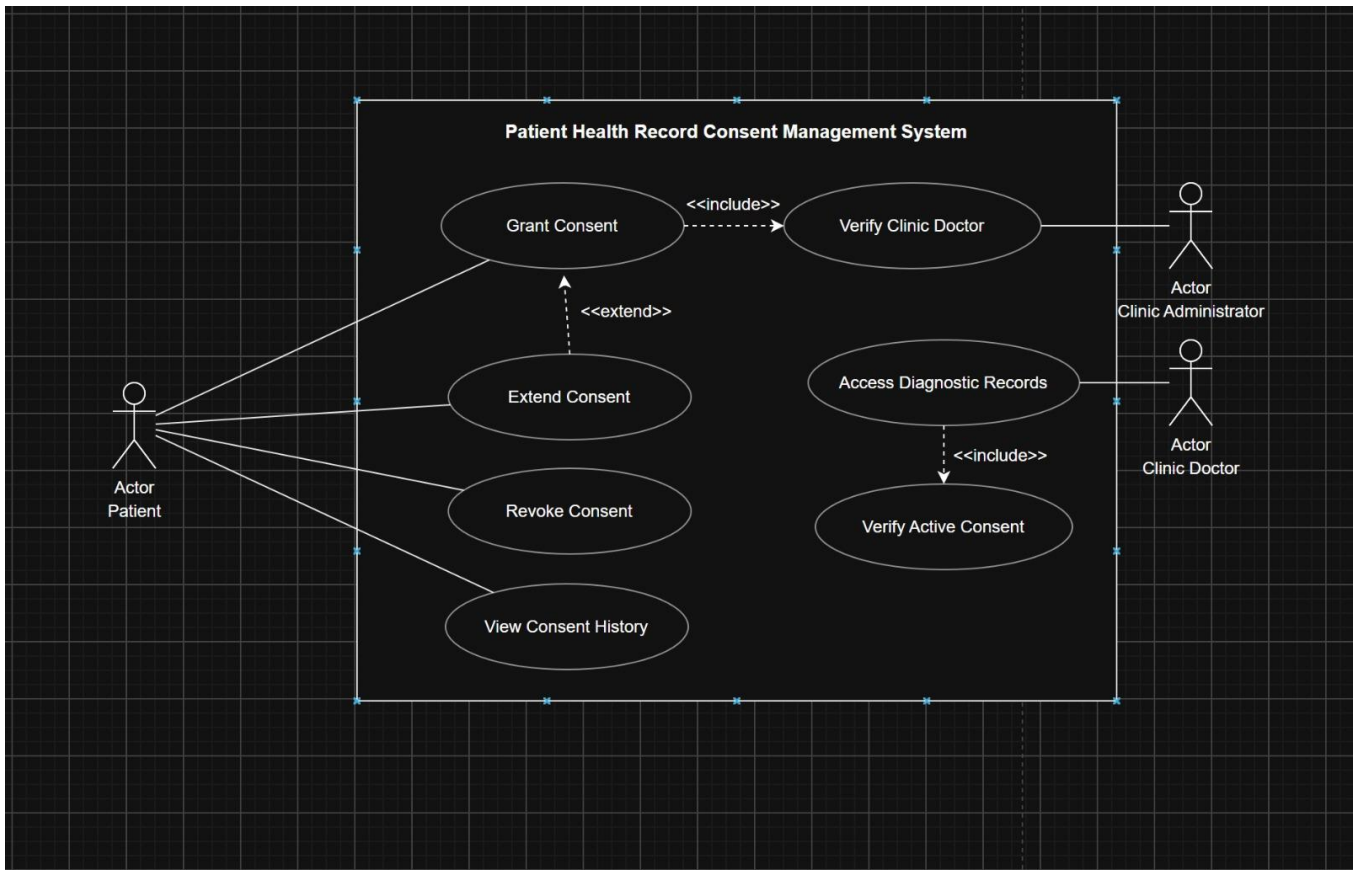
SRN: PES1UG24AM367

CLASS: AIML-G

1. Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-1	Functional	The system shall allow patients to grant time-bounded access permissions for specific diagnostic records to verified clinic doctors.	High	Pass: Consent request with a valid duration creates an active permission entry visible to the assigned doctor; Fail: No permission record is created or doctor cannot view the record.	Patients need direct control over who can see their medical data and for how long.
FR-2	Functional	The system shall allow patients to revoke an active consent permission before its expiry time.	High	Pass: Revoked permission is marked inactive immediately and doctor access is blocked on next attempt; Fail: Doctor can still access records after revocation.	Patients should be able to withdraw access at any time if they change their mind.
FR-3	Functional	The system shall allow a verified clinic doctor to access a patient's diagnostic records only when an active consent permission exists for those records.	High	Pass: Access request is granted only when a matching active consent record exists; Fail: Records are returned with no active consent or for a different patient.	Prevents unauthorized viewing of sensitive health records.
FR-4	Functional	The system shall automatically revoke access to consented records when the configured consent expiry time is reached.	High	Pass: Access attempt after the expiry timestamp is rejected without manual intervention; Fail: Doctor can still access records after expiry.	Ensures access does not outlive the patient's original permission window.
FR-5	Functional	The system shall allow patients to view the current status and history of their consent permissions.	Medium	Pass: Patient dashboard displays all past and active consents with their status and timestamps; Fail: Any consent action is missing from the displayed history.	Gives patients visibility into who has accessed or can access their data.
NFR-1	Security	The system shall permanently write all consent grant, revoke, and record-access events to an append-only audit trail.	High	Pass: Every grant, revoke, and access event has a corresponding audit log entry that cannot be edited or deleted; Fail: Any such event is missing from the log or the log can be modified.	An unchangeable audit trail is needed to prove compliance and investigate misuse.
NFR-2	Performance	The system shall process consent grant and revoke requests within 2 seconds for at least 95% of requests under 100 concurrent users.	Medium	Pass: Load test with 100 concurrent users shows 95th percentile response time ≤ 2 seconds; Fail: 95th percentile response time exceeds 2 seconds.	Consent actions often happen in urgent care situations, so slow responses could delay treatment.

2. UML Use-Case Diagram



3. Use-Case Flow Specification

Use Case Name	Grant Consent
Primary Actor	Patient
Preconditions	1. The patient is logged into the system. 2. The required diagnostic records are available. 3. The clinic doctor receiving access is verified.
Postconditions	1. The selected doctor can access the selected records. 2. Access remains valid only until the selected expiry time. 3. The consent action is recorded in the audit trail.
Main Success Scenario	1. Patient selects "Grant Consent". 2. System displays the available diagnostic records. 3. Patient selects the records to be shared. 4. Patient selects the verified clinic doctor. 5. Patient sets the consent expiry time. 6. System checks the doctor and consent details.

	7. System saves the consent. 8. System gives the doctor access to the selected records. 9. System records the consent action in the audit trail. 10. System shows a confirmation message. 11. The use case ends successfully.
Alternate Flow	6a. Invalid consent details: 1. System finds that the doctor is not verified or the expiry time is invalid. 2. System shows an error message. 3. The consent is not created. 4. Patient corrects the details. 5. The process continues from Step 3.